Wafer Yield as a Functional of a Marked Cox Process: Critical Area, Defect Clustering and Hierarchical Spatial Inference

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Abstract

We formulate die yield as the void probability of a marked point process of defects restricted to the critical region of a layout. Under a Poisson law this gives the exact model $Y = e^{-\lambda}$, where λ is the defect density times the size-averaged critical area, not the die area. Clustering is modelled by a Cox process: yield becomes a Laplace functional, which contains the Murphy, Seeds and negative binomial models as particular mixing laws and is never below the Poisson yield. We show that the clustering parameter is not intrinsic: it depends on the scale of the mixing relative to the die, and only large-scale mixing correlates neighbouring dies. A log-Gaussian Cox model with lot, wafer, radial and Matérn spatial effects plus a systematic component leads to a complementary log-log mixed model, derived rather than assumed. We give identifiability conditions, the law of good dies per wafer and a Bayesian inference procedure.

1. Defects as a marked point process

Let $W \subset \mathbb{R}^2$ be the wafer and $D_j \subset W$ the dies, of area A. Defects form a point process $\Phi = \{(X_i, R_i, L_i)\}$ on $W \times \mathbb{R}_+ \times \{1, \dots, m\}$ (position, diameter, layer). For a layout, a defect (x, r, l) kills die j iff $x \in C_{j,l}(r)$, the *critical region*; $A_{c,l}(r) = |C_{j,l}(r)|$ is the critical area [16, 10]. Let $K_j = \{(x, r, l) : x \in C_{j,l}(r)\}$. The defect-limited yield of die j is the void probability

$$Y_j = \mathbb{P}\big(\Phi(K_j) = 0\big), \tag{1}$$

Proposition 1 (Poisson yield)**.** *If Φ is Poisson with intensity $D_l(x) f_l(r)\, dx\, dr$ on layer l, the number of killer defects is $N_j = \Phi(K_j) \sim \mathrm{Pois}(\lambda_j)$ and $Y_j = e^{-\lambda_j}$, with*

$$\lambda_j = \sum_{l=1}^{m} \int_0^\infty \int_{C_{j,l}(r)} D_l(x)\, dx\, f_l(r)\, dr. \tag{2}$$

For uniform densities, $\lambda_j = \sum_l D_l \bar{A}_{c,l}$ with $\bar{A}_{c,l} = \int A_{c,l}(r) f_l(r) dr$.

Proof. The restriction of a Poisson process to a measurable set is Poisson with the restricted intensity [5]; $\Phi(K_j)$ is therefore Poisson with mean the intensity measure of K_j. □

The classical $e^{-D_0 A}$ [8] is thus the case $\bar{A}_c = A$; in general $\theta = \bar{A}_c / A \in [0, 1]$ is the probability that a defect landing in the die is fatal (the *kill ratio*), and the product form over layers is exact. Defect sizes follow the power law $f(r) = (p-1) r_0^{p-1} r^{-p}$, $r \geq r_0$, with $p \approx 3$ [19, 4].

Example. Two parallel lines of length L at spacing $s \geq r_0$, ignoring end effects: an extra-material defect of diameter r shorts them iff its centre lies in a strip of width $(r - s)_+$, so $A_c(r) = L(r - s)_+$. With $p = 3$,

$$\bar{A}_c = \int_s^\infty L(r-s) \frac{2 r_0^2}{r^3}\, dr = 2 L r_0^2 \Big(\frac{1}{s} - \frac{s}{2 s^2} \Big) = \frac{L r_0^2}{s},$$

and opens of a line of width w give $L r_0^2 / w$, so $\partial_s \log Y = D\, L r_0^2 / s^2$; for real layouts $A_c(r)$ is computed by Voronoi diagrams [13].

2. Clustering: Cox processes

Densities vary across lots, wafers and regions [17, 1]. Let Φ be Cox with random intensity measure Λ, i.e. Poisson given Λ [2]. Conditioning gives

$$Y_j = \mathbb{E}\big[e^{-\Lambda(K_j)}\big], \tag{3}$$

the Laplace functional of Λ at $\mathbf{1}_{K_j}$.

Proposition 2 (mixtures)**.** *(i) $Y_j \geq e^{-\mathbb{E}\Lambda(K_j)}$, with equality iff $\Lambda(K_j)$ is a.s. constant: clustering raises yield at fixed mean defect count. (ii) If $\Lambda = W\mu$ with $W \geq 0$, $\mathbb{E}W = 1$ and $\lambda = \mu(K_j)$, then $Y = \mathbb{E}e^{-\lambda W}$. $W \sim \mathrm{Gamma}(\alpha, \alpha)$ gives the negative binomial yield $Y = (1 + \lambda/\alpha)^{-\alpha}$ [19]; $W \sim \mathrm{Exp}(1)$ gives Seeds, $1/(1+\lambda)$ [15]; W triangular on $[0, 2]$ gives Murphy, $\big((1 - e^{-\lambda})/\lambda\big)^2$ [12]; $\alpha \to \infty$ recovers Poisson. (iii) $\mathrm{Var}\, N = \lambda + \lambda^2 \mathrm{Var}\, W$; for the gamma law $\alpha = 1/\mathrm{Var}\, W$, whose moment estimator is $\hat{\alpha} = \hat{\mu}^2/(\hat{\sigma}^2 - \hat{\mu})$.*

Proof. (i) Jensen for the strictly convex e^{-t}. (ii) The gamma Laplace transform is $(1 + t/\alpha)^{-\alpha}$; a triangular law on $[0, 2]$ is the law of $U_1 + U_2$ with U_i i.i.d. uniform on $[0, 1]$, whose Laplace transform is the square of $(1 - e^{-t})/t$. (iii) Law of total variance. □

The estimator $\hat{\mu}^2/\hat{\sigma}^2$ ignores the Poisson part of the variance; maximum likelihood on per-die counts is preferable.

Proposition 3 (scale of clustering)**.** *Let D_i, D_j be disjoint dies. (a)* Small-area clustering. *For a single defect size class with critical area A_c, if Λ is a gamma random measure on W with shape $\beta|\cdot|$ and rate b (independent increments), die outcomes are independent and $Y = (1 + 1/b)^{-\beta A_c}$: yield is log-linear in critical area, with effective density $\beta \log(1 + 1/b) < \beta/b = \mathbb{E}D$. (b)* Large-area clustering. *If $\Lambda = W\mu$, then $\mathrm{Cov}(\mathbf{1}_{\{N_i = 0\}}, \mathbf{1}_{\{N_j = 0\}}) = \mathrm{Cov}(e^{-\lambda_i W}, e^{-\lambda_j W}) \geq 0$, strictly unless W is degenerate; a block of k dies has yield $\mathbb{E}e^{-k\lambda W}$.*

Proof. (a) $\Lambda(C_j) \sim \mathrm{Gamma}(\beta A_c, b)$, with Laplace transform $(1 + 1/b)^{-\beta A_c}$ at 1, and increments on disjoint sets are independent. (b) Given W the counts are independent Poisson; both functions are non-increasing in W, hence positively correlated by Chebyshev's association inequality [3]. □

Hence α is not a constant of the process: it depends on the correlation length of Λ relative to the die, hence on area and block size [18, 7]. Windowing [8] identifies the Laplace transform of W only in regime (b).

3. A hierarchical spatial model

Let $Z_{\ell wj} \in \{0,1\}$ be the test outcome of die j (centre x_j) of wafer w in lot ℓ. We take a log-Gaussian Cox process [11]

$$\Lambda_{\ell w}(dx) = D_0\, e^{u_\ell + v_{\ell w} + h(\|x\|) + S_{\ell w}(x)} dx, \tag{4}$$

$\eta_{\ell w}(x)$ denoting the exponent, with $u_\ell \sim \mathcal{N}(0, \sigma_u^2)$, $v_{\ell w} \sim \mathcal{N}(0, \sigma_v^2)$, h a penalised radial spline (edge rings, centre spots) and $S_{\ell w}$ a Matérn field, written as the solution of $(\kappa^2 - \Delta)^{\alpha/2} S = \mathcal{W}$ so that it is a sparse Gaussian Markov random field on a mesh [9]. Losses that do not scale with critical area (mis-processing, parametric failures) enter as an independent systematic pass event with probability $Y_s(x) = \mathrm{logit}^{-1} g(x)$.

Proposition 4 (derived link)**.** *If $\eta_{\ell w}$ is constant on each die, then*

$$\mathbb{P}(Z_{\ell wj} = 1 \mid \eta) = Y_s(x_j) \exp\big(- D_0 \bar{A}_c\, e^{\eta_{\ell w}(x_j)} \big). \tag{5}$$

For $Y_s \equiv 1$ this is a Bernoulli mixed model for the failure event with complementary log-log link, $\log\big(- \log \mathbb{P}(Z = 1 \mid \eta)\big) = \log \bar{A}_c + \log D_0 + \eta(x_j)$*, and offset* $\log \bar{A}_c$.

Proof. Proposition 1 given η, and independence. □

The link follows from Poisson thinning; when η varies within a die, $e^{\eta(x_j)}$ is replaced by the die average $A^{-1} \int_{D_j} e^{\eta}$, a linear functional of the field on the mesh. The marginal yield $Y_j = Y_s(x_j) \mathbb{E} \exp(-D_0 \bar{A}_c e^{\eta_j})$, $\eta_j \sim \mathcal{N}(m_j, \tau_j^2)$, is the Laplace transform of a log-normal variable; it has no closed form and is computed by Gauss–Hermite quadrature, $\pi^{-1/2} \sum_k w_k \exp(-D_0 \bar{A}_c e^{m_j + \sqrt{2}\tau_j t_k})$.

Proposition 5 (identifiability)**.** *From one critical area, only $Y_s e^{-D_0 \bar{A}_c}$ (Poisson) or $Y_s (1 + D_0 \bar{A}_c/\alpha)^{-\alpha}$ (gamma) is identified. With distinct critical areas $a_1 < \cdots < a_K$ (windowing in regime (b), or several products on the same process), (Y_s, D_0) are identified for $K \geq 2$ in the Poisson model, and (Y_s, D_0, α) are locally identified for $K \geq 3$ outside a Lebesgue-null set of designs.*

Proof. Poisson: $\log Y(a) = \log Y_s - D_0 a$ is affine. Gamma: with $t = D_0/\alpha$, the derivatives of $\log Y(a)$ with respect to $(\log Y_s, D_0, \alpha)$ are 1, $-a/(1+ta)$ and $-\log(1+ta) + ta/(1+ta)$, three linearly independent analytic functions of a; the 3×3 Jacobian determinant is a non-zero analytic function of (a_1, a_2, a_3), so it vanishes only on a null set. □

4. Wafer yield, redundancy and inference

Good dies per wafer. $G_w = \sum_j Z_{wj}$ has $\mathbb{E} G_w = \sum_j Y_j$ and $\mathrm{Var}\, G_w = \sum_j Y_j(1 - Y_j) + \sum_{i \neq j} \mathrm{Cov}(Z_i, Z_j)$, which exceeds the binomial variance under Proposition 3(b). Its posterior predictive law, simulated from (u, v, S) then Z, is the model-based wafer yield histogram; excursions (its left tail) are a two-state mixture on $v_{\ell w}$ whose posterior state probability is an SPC-type alarm.

Defect tolerance. If redundancy repairs up to k_0 killer faults [6], $Y = \mathbb{P}(N \leq k_0)$; under Proposition 2(ii), with $q = \lambda/(\alpha + \lambda)$,

$$Y = \sum_{k=0}^{k_0} \frac{\Gamma(\alpha + k)}{\Gamma(\alpha)\, k!} (1 - q)^\alpha q^k, \tag{6}$$

with the block-level law of [7] when spares are shared.

Joint inference. In-line inspection gives the positions of detected defects on layer l, a thinned version of Φ with detection probability π_l. Modelling these point patterns as a log-Gaussian Cox process sharing the field $S_{\ell w}$ with the test data couples inspection and electrical test in a single likelihood; the kill ratios $\theta_l = \bar{A}_{c,l}/A$ are then estimated rather than assumed. Without the systematic mixture the model is latent Gaussian and is fitted by integrated nested Laplace approximations [14]; with it, by Hamiltonian Monte Carlo with the discrete states marginalised.

Validation. Models (Poisson, gamma, LGCP, with or without h, S) are compared by leave-one-wafer-out cross-validation, which respects within-wafer dependence, with the log score of G_w and PIT calibration.

5. Conclusion

Classical yield models are Laplace transforms of particular mixing laws, and the clustering parameter reflects a scale rather than a constant. A log-Gaussian Cox model with a systematic component gives an identifiable complementary log-log spatial model, from which die, wafer and repaired yields follow with their uncertainty.

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